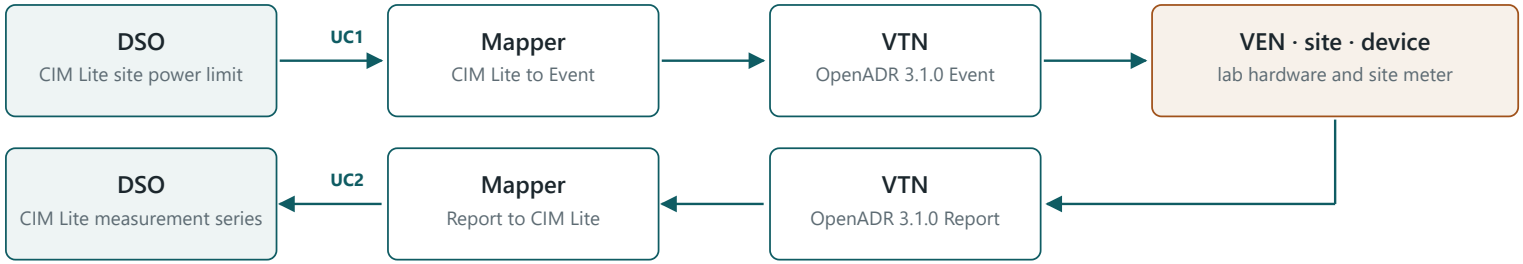
CIM–OpenADR mapping: December 2026 lab test

Site power limits and measurement reports

ElaadNL plans to test two mappings with lab hardware in December 2026. UC1 converts a DSO site power limit into an OpenADR 3.1.0 Event. UC2 converts the VEN's measurement Report into a CIM Lite measurement series.

The first test covers maximum import limits and interval-average import power, using existing OpenADR payloads. The site boundary, measurement method and response to faults must be agreed before the run (page 6).



VTN the OpenADR server that publishes Events and receives Reports. **VEN** the OpenADR client at the site, connected to the site controller and devices. **CIM Lite** this project's proposed JSON application profile, based on IEC CIM. The test uses that JSON format rather than IEC XML.

Test objectives

- The VEN receives a site import limit as `IMPORT_CAPACITY_LIMIT`. Device logs and an independent meter show whether the device applied it.
- The VEN reports the mean import power per Event interval (`DEMAND`, `MEAN`), linked to the Event.
- Check which limit the device applies after a scheduled limit ends, then test connection loss separately.

Outside the test scope

- Mandatory-flow minimum limits, export reports, dispatch and flexibility (pages 4 and 7).
- Interoperability beyond the lab partners, and certification.
- Settlement-grade measurement: Reports are operational telemetry.
- An interval mean cannot show that the limit was enforced at every moment.

Worked example

7.0 kW maximum import · **6.4 kW** reported mean

16:00–17:00 UTC on 2026-10-16. The following mappings use this example data; hardware results are still to come.

UC1 — Map the site power limit to an Event

Input `examples/operating-envelope-cim.json`; output: the body of `POST /events`. The table shortens the instruction identifier to `‹id›` and the connection-point identifier to `‹EAN›`. The example target is `cp:A10:871999999999999990`. Full field paths are retained in the HTML tooltips.

INFORMATION	CIM LITE	OPENADR EVENT	RULE
Instruction identity	mRID **`‹id›`** revisionNumber **`1`**	eventName **`cim-lite: OperatingEnvelope:‹id›:1`**	The Event name carries the prefix, mRID and revision by profile convention. The VTN assigns the Event id. CL-09 · R-ID-2
Connection point	codingScheme **`A10`** value **`‹EAN›`**	target **`cp:A10:‹EAN›`**	Profile convention: cp:<scheme>:<value>. The same string is the Report target and resourceName. CL-02 · R-ID-1
Start time	start **`2026-10-16T16:00Z`**	start **`2026-10-16T16:00:00Z`**	Same instant: the ESMP minute timestamp becomes RFC 3339 UTC. CL-05 · R-TIME-1
Interval length	resolution **`PT1H`** curveType **`A01`**	duration **`PT1H`**	Fixed-size blocks (A01): the resolution becomes the default interval duration. CL-05 · R-TIME-3
Interval number	position **`1`**	id **`1`**	Profile convention: the point position becomes the interval id. OpenADR ids are not sequence numbers. CL-05 · R-ID-3
Limit type	limitNature **`hardLimit`** flowConvention **`load`**	payload type **`IMPORT_CAPACITY_LIMIT`**	A hard limit on import-positive power becomes a maximum import. Setpoints and relative values are rejected. CL-03 · CL-08 · CL-10 · R-ENV-1
Limit value	value **`7`**	value **`7.0`**	The value after unit conversion: 7 KWT = 7.0 kW. CL-03 · R-ENV-4
Unit	unit **`KWT`**	units **`KW`**	KWT becomes KW (WTT ×0.001, MAW ×1000). W with a multiplier is also accepted; Wh is energy and is rejected for a power limit. CL-07 · R-UNIT-1
Export limit	lowerLimit *absent*	*not sent*	No lower bound, so no EXPORT_CAPACITY_LIMIT is sent. Absent is not zero. CL-04 · R-ENV-3
Measurement request	*mapper option*	payloadType **`DEMAND`** readingType **`MEAN`**	The Event asks the VEN for the mean power in kW per Event interval, for the same target. CL-11 · R-OBS-1

UC2 — Map the measurement Report to CIM Lite

Input `examples/observation-report.json`, the Report for Event `evt-20261016-0001`. The mapper checks the Event id, the target, the interval id and the interval times before it writes a CIM Lite measurement series. `‹id›` and `‹EAN›` are as on page 2.

The reported mean is 6.4 kW, below the 7.0 kW import limit. A separate meter trace is needed to assess excursions within the interval.

INFORMATION	OPENADR REPORT	CIM LITE	RULE
Event link	eventID **`evt-20261016-0001`**	instructionRef **`‹id›`**	The Report's eventID must equal the Event id. The CIM mRID is read back from the Event name. CL-15 · R-OBS-5
Connection point	resourceName **`cp:A10:‹EAN›`**	codingScheme **`A10`** value **`‹EAN›`**	Must be one of the Event targets; otherwise the Report is rejected. CL-02 · R-ID-1
Measurement type	payloadType **`DEMAND`** readingType **`MEAN`**	measurand **`ActivePower`** aggregation **`average`**	DEMAND with MEAN is the interval-mean active power. It describes the averaging, not how the meter measured it. CL-11 · R-OBS-4
Interval	start **`2026-10-16T16:00:00Z`** duration **`PT1H`**	start **`2026-10-16T16:00Z`** resolution **`PT1H`**	The interval id and its exact times must match the Event interval; otherwise the Report is rejected. CL-05 · R-OBS-5
Measured value	value **`6.4`**	value **`6.4`**	The interval mean is copied unchanged. CL-11 · R-OBS-6
Unit	units **`KW`**	unit **`KWT`**	DEMAND must be in kW; W is rejected by the profile. Written as KWT. CL-07 · R-UNIT-3
Quality	*DATA_QUALITY absent*	quality **`A04`**	Missing DATA_QUALITY maps to ENTSO-E A04 ('as provided'), with no assertion of validity. CL-13 · R-OBS-3
Evidence grade	*profile default*	evidenceGrade **`operationalTelemetry`**	OpenADR cannot mark data as settlement-grade, so it is recorded as operational telemetry. CL-16 · G-07

Rejected inputs: an unrelated target or interval, an export mean, or units other than KW. CIM-to-Report conversion also rejects quality codes without a faithful OpenADR equivalent. Mapping limitations: page 8.

Requirements and gaps for the lab test

Use these requirements to configure the VTN and VEN and prepare the test cases. The first test uses existing OpenADR payloads and local agreements. Related proposals are listed on page 8.

ISSUE	REQUIREMENT FOR THIS TEST	HOW TO CHECK IT
L-01 Connection-point identity G-03 · G-11 · CL-02 · UC1 UC2	VTN and VEN use the same target string, cp:A10:<EAN-18>, and the Report's resourceName repeats it. The VTN grants this target to the VEN. A lab registry links it to the physical site meter and rules out virtual or aggregated points.	A Report with a different target is rejected (CORR-TARGET). Confirm that the VEN can read only the Events for its own target.
L-02 Instruction reference G-10 · E-09 · CL-09 · CL-15 · UC1 UC2	The Event name carries the CIM instruction: cim-lite:OperatingEnvelope:<mRID>:<revision>. The VEN only has to quote the Event id in its Report. The VTN keeps a history of Event versions.	Map a Report back to the CIM instruction through its eventID. Send a late Report after an Event update and check that it maps to the version that was active.
L-03 Site limit, not device limit G-05 · UC1 UC2	Import and export limits apply to the whole site at the grid connection. The site meter is the measurement point, and uncontrolled lab loads count towards the limit.	List every load and generator behind the site meter before the run.
L-04 Export measurements G-06 · E-05 · CL-11 · UC2	The first test covers import only. DEMAND cannot be negative, so net export cannot be reported and a negative mean is rejected (OBS-NEGATIVE). A positive mean can still hide export within the interval.	Switch off PV or battery export, or detect it, and mark any run with export as outside the test scope.
L-05 Minimum (mandatory-flow) limits G-20 · E-01 · CL-03 · CL-04 · UC1	Only maximum limits are sent. A lower bound of 0 or less becomes an export limit. An envelope that forces import (lower bound > 0) or export (upper bound < 0) is rejected (ENV-MANDATORY-*).	Submit one mandatory-flow envelope and confirm that the mapper rejects it.
L-06 Schedule expiry and fallback G-20 · G-23 · E-02 · TX-08 · UC1	Preload a long, lower-priority fallback Event. Agree what applies after the fallback ends (*_CAPACITY_SUBSCRIPTION or the TR 62746-2 unlimited rule) and after a VEN restart. A fallback Event does not override an active higher-priority Event when the connection is lost; that case is L-07.	Let a scheduled Event end and check that the device moves to the fallback limit. Restart the VEN and check which limit it applies.
L-07 Connection loss G-19 · E-03 · UC1	Agree how the VEN checks that the VTN is reachable, the timeout for connection loss, the failsafe limit and the recovery procedure. A successful poll can confirm reachability; silence alone cannot. Record the poll interval.	Cut the network during an active Event and measure the time to failsafe and to recovery. Test this separately from schedule expiry (L-06).

Requirements and gaps for the lab test (continued)

ISSUE	REQUIREMENT FOR THIS TEST	HOW TO CHECK IT
L-08 Device execution G-19 · E-04 · UC1	Check command acceptance and physical response separately. The VTN's HTTP response confirms Event storage. A reported mean below the limit does not establish that the device applied the command.	Record when the VEN receives the Event, when the device accepts or rejects the command, and when the limit takes effect. Run a load that would exceed the limit without control. Check the response using the independent meter and the agreed response-time limit.
L-09 Meter intervals G-21 · CL-11 · R-OBS-1 · UC2	The VEN reports the mean import power per Event interval (DEMAND, MEAN). Agree whether the meter reports net or import-only power, how the mean is calculated, and how incomplete intervals are flagged. For IEC 61968-9 input, the timestamp marks the interval end: 17:00 with a one-hour duration represents 16:00–17:00 UTC.	Compare the reported mean with the independent meter for each interval, including the first and last interval and one after a restart. Also agree the sampling rate, the coverage threshold for a valid mean, and how register rollover and reset are handled.
L-10 Missing or estimated values G-12 · E-06 · CL-13 · CL-14 · UC2	A missing interval is sent as DEMAND 0 with DATA_QUALITY MISSING, so a receiver that ignores DATA_QUALITY reads 0 kW. Missing, estimated or bad values never count as evidence of compliance.	Interrupt the meter for one interval. Check that MISSING is sent and maps back to 'no value' (ENTSO-E A02).
L-11 Errors in the OpenADR examples G-13 · E-12 · UC1 UC2	The User Guide's 'start now' date 0001-01-01 is invalid, and its envelope example lacks objectType. Use explicit UTC start times and validate against the pinned 3.1.0 schema, not the Guide examples.	Validate every message in the run against the schema and enumerations.
L-12 Overlapping Events R-CONFLICT-2 · E-02 · UC1	OpenADR does not say which of two overlapping Events with equal priority applies. Give the fallback a different priority. If a tie still occurs, the most restrictive limit applies.	Test overlap, update and cancellation of Events.
L-13 Limit type and obligation G-04 · E-13 · CL-10 · UC1	Record that the command specifies a maximum at the site boundary and that compliance is binding. Treat limit type and obligation as separate settings in the test plan.	Record both agreements in the test plan.
L-14 Response time and pass/fail criteria E-10 · TX-34 · UC1 UC2	Agree the response time, settling time, tolerance and pass/fail windows. Interval means cannot show instantaneous enforcement.	Judge each run against these values using the independent meter trace.

Agree before testing

Record the agreed settings, acceptance criteria and responsible person before the run.

SETTING	WHAT TO AGREE	RELATED	AGREED VALUE · OWNER
Site boundary	Which meter forms the site boundary, and which loads and generators sit behind it.	L-01 · L-03	
Measured quantity	Net or import-only power; how the interval mean is calculated; the coverage threshold; register rollover and reset; whether timestamps mark the interval start or end.	L-09	
Clock alignment	The maximum allowed clock offset for the VTN, VEN and meter. Record the measured offsets before the run and after a restart.	—	
Event delivery	The poll interval, and the lead time between publishing an Event and its start. Allow for the poll phase, network retries, VEN processing, device response and clock offset. Include one late-arriving Event.	L-07	
Report interval	When the VEN sends Reports. The example data asks for one Report at the end of the last interval, which is too late to follow a one-hour test live.	L-09	
Device response	Response time, settling time, tolerance and pass/fail windows.	L-08 · L-14	
Connection loss	Timeout, failsafe limit and recovery procedure.	L-07	
After fallback and restart	Which limit applies when the fallback Event ends, and after a VEN restart.	L-06	
Limit type and obligation	Hard site maximum, not a setpoint; binding.	L-13	
Units	kW for limits and DEMAND. A value in W is schema-valid but rejected by the profile.	—	
Test times and logging	Generate future start times when the test runs; the example data uses October 2026. Log publication, fetch, command acceptance, applied limit, meter interval and Report arrival separately.	—	
Event lifecycle cases	Which of these to run: late, duplicate and out-of-order Events; update; cancellation; VEN reboot; stale queued commands.	L-12	

Gaps outside the first test

The work below is outside the first test. References to L-rows identify the local agreement used where needed. Proposed additions are this project's intended solutions; they are not part of OpenADR 3.1.0. A profile rule uses existing OpenADR objects. Open items need a decision or further standards work.

ISSUE	WHY IT IS OUTSIDE THIS TEST	REMEDY STATUS	REFERENCES
Flexibility and commercial exchange			
N-04 Baseline and reference-curve role	Belongs to the flexibility use case (UC3).	Open	G-08 · CL-17 · CL-18
N-05 Two-sided relative flexibility band	Belongs to UC3 and the commercial track.	Open	G-09 · CL-19
N-06 Bids, offers and settlement	Commercial track, separate interface.	Open	G-16
N-03 Settlement-grade evidence	Lab data is operational telemetry; settlement uses a separate metering chain.	Open	G-07 · CL-16
N-15 Prices and tariffs	Other use case. PRICE and EXPORT_PRICE already exist in 3.1.0.	In OpenADR 3.1.0	TX-26 · TX-27
Dispatch and control extensions			
N-09 Signed or directional dispatch	The first test sends limits, not setpoints.	Proposed addition	G-22 · E-07
N-10 Release to normal operation	Needed for dispatch. Ordinary limit expiry is tested in the first test. First test: see L-06.	Proposed addition	G-23 · E-08
N-11 Resource capability attributes (ramp, response time)	A standard attribute is outside this test scope; the response time is agreed locally. First test: see L-14.	Open	E-10 · TX-34
N-13 Binding vs advisory carrier	A wire field is outside this test scope; the obligation is agreed locally. First test: see L-13.	Open	E-13
Metering, profile and source issues			
N-01 Envelope roles and limit type in CIM (IEC 62746-4)	Needs a change in the IEC standard (TC57 WG21). The test uses a local agreement. First test: see L-13.	Open	G-04 · CL-10
N-02 CIM release and enumeration drift	Affects DSO-side CIM input (INSIEME/ESMP vs IEC 62746-4). The test uses the agreed CIM Lite version.	Profile rule	G-02
N-07 Energy (USAGE) reporting	The test reports mean power, not energy. First test: see L-09.	Open	G-14 · CL-12
N-08 Measurement quality and V/A/var units in IEC 62746-4	Only relevant when the DSO side uses IEC 62746-4 XML.	Open	G-17 · G-18
N-12 Frequency and per-phase readings	The test reports mean active power only.	Proposed addition	E-11 · G-21
N-14 Unit missing in EDDIE RECMMOE	Third-party format, not used in the test.	Open	G-15

Proposals and implementation checks

These proposals and profile rules relate to the test requirements. New payloads are outside the first test. The optional trials require a private agreement and are not implemented by the mapper.

REQUEST	TOPIC	STATUS	ISSUE	OPTIONAL TRIAL
E-09	External instruction reference (EXTERNAL_REFERENCE payload)	Proposed addition	L-02	—
E-05	Export interval measurements (EXPORT_DEMAND / EXPORT_USAGE)	Proposed addition	L-04	EXPORT_DEMAND as an agreed private payload
E-01	Minimum import / export capacity	Proposed addition	L-05	IMPORT/EXPORT_CAPACITY_MINIMUM as agreed private payloads
E-02	Fallback profile (lower-priority overlapping Events)	Profile rule	L-06	—
E-03	Heartbeat and failsafe	Proposed addition	L-07	HEARTBEAT report (private enumeration value)
E-04	Execution acknowledgement report	Proposed addition	L-08	EVENT_RESPONSE report straight after the Event (private enumeration value)
E-06	Additional DATA_QUALITY values	Proposed addition	L-10	—
E-12	Errata in the 3.1.0 material	Editorial corrections	L-11	—
E-13	Binding vs advisory instruction	Open	L-13	—
E-10	Resource capability attributes (ramp, response time)	Open	L-14	—

UC1: context and metadata

The VTN/program configuration supplies the sender identity. The revision is carried in the Event name; limit meaning follows from the payload and L-13. The VTN adds `id`, `createdDateTime`, `modificationDateTime` and `objectType`. Reports quote its Event id.

UC2: measurement limitations

The mapping between OpenADR and ENTSO-E quality codes loses detail. The Report has no settlement evidence grade; this profile records operational telemetry. Device execution and excursions within the interval need separate logs and a meter trace.

Validation and sources

The example envelope was converted to an Event and back without changing the model values (check passed); the sender identity was supplied from configuration. The example Event and Report pass the schema, enumeration and profile checks. The mapper's compliance status for the example interval is `WITHIN_ENVELOPE_MEAN`, which covers the interval mean only. Hardware testing is pending.

Technical reference: field paths, evidence levels and concept classifications per row are in `docs/04-mapping-matrix.md`; all gaps in `docs/07`, OpenADR change requests in `docs/10`, sources in `docs/02`. Sources: OpenADR 3.1.0 (openadr3.yaml and enumerations, pinned); IEC 62746-4:2024 with Annex C XSDs; IEC TR 62746-2:2025; IEC 62325-351, 62325-301, 61970-301, 61968-9 and 61968-11; ENTSO-E code lists. Generated by `python -m cim_openadr build-brief` from the mapper, `examples/` and `mappings/`.